Interdisciplinary Collaborative Core (ICC)

CC0006: Sustainability: Society, Economy, and Environment
Semester 1, AY 2024/25

Section 1: Written Report

The written report is a team assessment, and accounts for 25% of your grade. Each team is to hand in a written report according to given instructions. The report should not be more than 2000 words with a 10% leeway. Sections not included in the word count: (i) cover page, abstract, content page, and references. In principle, all team members will receive the same mark. However, if the evaluation of team members' contribution to teamwork (see appendix 3) shows that the contribution of a team member is significantly different to that of the others, the individual grade may be adjusted to reflect that.

Broad categories of different project types that you can adopt (templates will be provided):

- **Literature study:** Propose a new and innovative method of solving a sustainability-related problem: provide a detailed theoretical analysis based on relevant and credible sources.
- **Case study/global best practices-based project:** Analyse an existing solution (technology or policy-based), add your own analysis of its feasibility in the context of Singapore or a broader local context (i.e., Southeast Asian region), and draw conclusions.
- **Survey:** Short-survey-based projects, online or in-person surveys.
- **Site visit-based projects:** A site visit to clean energy buildings and analysis of the associated technology/policy approaches, e.g., visit Singapore's zero-energy building, make a detailed report on its features and your understanding of it, add onsite clicked photographs (follow rules of onsite photography), add an analysis section on the feasibility of more of such buildings, compare it with other such buildings across the world, etc.

Section 2: Presentation Session

The video presentation is a team assessment and accounts for 25% of your grades. For this assessment, you are to record a 12-15 min (maximum 15 mins) team presentation on video and submit it to your instructor through NTULearn. Other students will watch your video presentation and give qualitative feedback through the Question and Answer (Q&A) session conducted in week 13 at a specified venue.

Presentation Q&A Session Format

- Instructors will inform you of the order of presentations.
- Presenting group to provide us with a 2-3 mins (maximum) recap of your group topic before we begin to take questions from the rest.
- Venue and location will be provided later.

Section 3: Academic Integrity:

Good academic work depends on honesty and ethical behaviour. The quality of your work as a student relies on adhering to the principles of academic integrity and to the NTU Honour Code, a set of values shared by the whole university community. Truth, Trust and Justice are at the core of NTU's shared values.

As a student, it is important that you recognize your responsibilities in understanding and applying the principles of academic integrity in all the work you do at NTU. Not knowing what is involved in maintaining academic integrity does not excuse academic dishonesty. You need to actively equip yourself with strategies to avoid all forms of academic dishonesty, including plagiarism, academic fraud, collusion and cheating. If you are uncertain of the definitions of any of these terms, you should go to the academic integrity website for more information.

On the use of technological tools (such as Generative AI tools), different courses / assignments have different intended learning outcomes. Students should refer to the specific assignment instructions on their use and requirements and/or consult your instructors on how you can use these tools to help your learning.

Section 4: Potential Topics for Group Projects

We will be providing a list of potential topic suggestions for your upcoming group projects. Your group should still formulate your own title, research question, and focus. You are expected to develop a unique approach to address the topic of your choice. You may also choose your own topic, as long as your cluster instructor approves it.

Section 4.1: Group Project Topics (Sustainability Office Problem Statements)

	Problem Statement
1.	Carbon & Energy: • What are some best practices that you can undertake to reduce energy consumption in laboratories? • What are some behavioral measures you can undertake to reduce your overall carbon footprint around campus? • How can we encourage students staying in halls to ensure the effective and responsible use of air conditioning?
2.	Waste Management & Recycling: • How might we encourage the NTU population to bring their own containers and bottles for takeaways on campus? • How should we handle the food waste generated in canteens across the NTU campus? • What are some successful examples of upcycling projects from other universities or institutions and how can it be applied in NTU campus?
3.	Water Conservation & Management: • How can we ensure effective operational strategies for minimizing water losses due to leaks? • How can the university encourage faculty and staff to adopt water-saving behaviors in their daily routines? • How can the university optimize water use in its landscaping and outdoor spaces?
4.	Food Security & Sustainable Food Production: • With the recent approval of certain insect species for human consumption, do you think there will be a high adoption rate of insect-based foods on campus?

	• How can we minimize food waste on campus, particularly in canteens and halls? • How can we encourage the NTU community to select more plant-based food options?
5.	Responsible Consumption & Circular Economy: • Certified sustainable cooking oil ensures that the cooking oil (palm oil) is ethically produced without deforestation, how might we encourage more vendors in NTU to adopt the use of sustainable cooking oil? • How might we reduce our reliance on single-use disposables on campus? • How can we encourage enterprises run on campus, such as bookstores, cafeterias, food stalls, and service centers, to adopt and promote sustainable practices that align with NTU's sustainability goals?
6.	Sustainable Urban Solutions: • How can NTU develop and/or promote sustainable transportation options to reduce the environmental impact of commuting students and staff? • How can we ensure the comfort of facilities for users while ensuring sustainability? • How can we improve on existing infrastructure on campus to ensure better accessibility to the wider NTU community?
7.	Sustainable Education: • How can NTU promote interdisciplinary collaboration among students and faculty to address complex sustainability challenges through innovative projects and research?

	• What methods can NTU use to assess the sustainability literacy of their students, staff, and faculty, and how can we improve sustainability education outcomes based on these assessments? • How can NTU empower students who are passionate about sustainability to apply their knowledge and expertise to make NTU more sustainable?
8.	Biodiversity & Conservation: • How can we better ensure that the NTU community can co-exist peacefully with wildlife on campus? • What are some of the interventions that NTU can consider preventing roadkill from happening? • How can we integrate nature and biodiversity into the halls to ensure students' access to green spaces?

Section 5.2: Group Project Topics (Open Categories)

	Topic
1.	**Green Energy**
	Fuel • Sustainable Aviation Fuel • Future of fuel cell/hydrogen fuel-based vehicles in Singapore **Housing** • Materials for energy harvesting in buildings. • Zero energy building in Singapore **Energy Generation** • Solar energy: Large-scale implementation of photovoltaic cells. The potential of solar cells as an energy source for Singapore. • Innovative green energy sources from NTU.

	• Renewable options for Singapore • Rooftop solar PV deployment in Singapore: How to accelerate the adoption. • Thermoelectric generation from waste heat. • Economic, social, and environmental impacts on clean and renewable energy. • A new generation of nuclear innovation for a greener future.
2.	**Waste and Recycling**
	• Recycling of EV battery materials • Sustainable waste management technologies in Singapore, NTU or examples from around the globe. • Is the pretext of recycling leading to more waste: especially plastic and textile waste? • Recycling in Singapore: current shortcomings and potential solutions. • Closing the various waste loops in Singapore. • E-waste management: suggestions and recommendations in Singapore? • Food composting and assessing the present capability in Singapore, e.g. food composting plant model to be used in NTU. • Food packaging problem - strategies and barriers to reducing packaging.
3.	**Pollution**
	• Bioremediation of pollutants, see: https://www.sciencedirect.com/topics/earth-and-planetary-sciences/bioremediation. • Suitability of different sensors for detection of water pollution in Singapore. • The social, economic, and environmental cost of haze for Singapore. • How is chemical pollution making rainwater on earth unsafe to drink? • Sustainable municipal food waste management in Singapore. • Microplastic are everywhere: occurrence, perils, and eradication. • Light pollution affects global biodiversity and human well-being, see: https://www.sciencedaily.com/releases/2021/09/210914111302.htm. • Noise pollution of oceans and rivers – conflicting interests of human development and wildlife, see:

	https://www.forbes.com/sites/evaamsen/2021/02/21/human-noise-is-taking-over-the-ocean-soundscape/?sh=37faf901241b.
4.	**Circular Economy**
	• Is the digital economy a step towards or away from sustainability? • Willingness to pay for the environment: this can be a multi-country comparison with the baseline or focus on Singapore. • What are some initiatives Singapore has undertaken to ensure a circular economy? • Is the pretext of green and organic products leading to more consumption – survey-based project. • Circular economies related to: Responsible design, production, distribution, consumption, product-life cycles, and upcycling. See: https://www.wired.co.uk/article/supply-chain-oversupply; https://www.bbc.com/news/business-44885983.
5.	**Food Security and Sustainable Food Production**
	• Future Ready Food Safety Hub (FRESH) Programme: Building local food safety capabilities to support growing innovation in food production and manufacturing and developing new food safety standards. • Hydroponic Rooftop Farming in Singapore: A masterstroke or futile attempt? • Improving the food security status among Singaporean households. • Urban farming in Singapore: environmental, economic, and social perspectives • Recent advances in consumer preferences and behaviour toward healthy and functional foods. • Saving the planet by eating - the role of food and diets for a sustainable global food system and future food security (can focus on meat/veg, or local/imported foods, or future sustainable agriculture (including Singapore urban farming initiatives). • Food Security and Climate Change in ASEAN. • How can we build a Singapore culture of sustainability of low or no wastage of food? • Are insects the future of food?

6.	**Mitigating and Adapting to Climate Change**
	• How could NTU campus lower its carbon footprint? NTU aims to halve net energy utilization, water usage, and waste generation each by March 2026, compared to the levels of 2011. • Sea level rise - how does it work and what will be the effects in Singapore/Southeast Asia? What precautions is Singapore taking? • Sustainable coastal protection strategies to combat rising sea levels. • Natural hazard monitoring and disaster response. • Carbon capture technology paves the way for closing the carbon cycle. • Natural Climate Solutions - what are they and how can they help Singapore? • Palm oil and the carbon cycle - links between agriculture and CO2 emissions. • ASEAN policies to prevent and mitigate climate change in the region. • How blue carbon and/or green carbon can help mitigate climate change.
7.	**Water - Sustainable Supply and Usage**
	• Sustainable water usage and water supply. • Energy-efficient wastewater treatment processes. • Urban sewage systems as mines for resource mining. • Emerging contaminants in the aquatic environment. • Is the world running out of fresh water? See: https://www.bbc.com/future/article/20170412-is-the-world-running-out-of-fresh-water. • How can Singapore be self-sustainable in meeting its water needs and have a stable water supply?
8.	**Responsible Consumption and Production**
	• Environmental impact of the fashion industry. • Sustainable fashion initiatives from around the world. • Innovative alternatives to plastic bags.
9.	**Social Equality and Inclusion**

	• Innovative ways to promote social inclusion and equality in Singapore: explore policy methods, social marketing, or purely community-based approaches. • Improving the lives of low-income and vulnerable people in Singapore. • Promote awareness of gender discrimination in the workplace. • Income inequality in Singapore: causes, consequences, and mitigations • Sustainable communities and neighbourhoods: possible opportunities for environmental, social, and economic development. • Climate change, water scarcity, and gender: perspectives on human vulnerability to climate change effects. • What can we learn from China’s successes in lifting 700 million people out of poverty?
10.	**Sustainability Psychology**
	• What makes Singaporeans happy: how long do consumption-based happiness last and how does it impact society’s move toward sustainable consumption and well-being? • Is the world leaving its disadvantaged behind: explore national plans (if any) to support low-skilled workers in the upcoming world of AI and robotics. • Is the pretext of green and organic products leading to more consumption: this could be a survey-based project.
11.	**Sustainable Cities**
	• Economic and social feasibility of cycling districts in Singapore: compare it with the best practices around the world. • Story of the resilience of communities: Find one in Singapore, if not broaden your geographical boundary: Southeast Asia - Asia - Eurasia - etc. • Smart infrastructure: an emerging frontier for smart cities. • How to accelerate EV adoption for a more sustainable land transport system? • Protecting urban biodiversity: A call to action. • What is Socio-ecological sustainability?

	• How do population and economic growth along with transforming cities encroach into natural and indigenous ecologies? What becomes of new priorities, and balance to be addressed, for definitions of sustainability? See: https://www.sciencedirect.com/topics/social-sciences/socio-ecological-system.
12.	**Health and Sustainability**
	• Strategies for prevention of the next pandemic • A closer look behind UN Sustainable Development Goal 3: Ensure healthy lives and promote well-being for all at all ages. See: https://sdgs.un.org/goals/goal3.
13.	**Sustainability Education**
	• Sustainability and Universities: How to reach the SDG(s) through education. See: https://sdgs.un.org/goals/goal4.
14.	**Sustainability and Businesses**
	• How does environmental, social, and governance (ESG) relate to Sustainable Development Goals? • How's the landscape of Singapore companies' sustainability reporting performance?
15.	**Sustainability and the Uncertain Future**
	• Will food/water crises lead to future conflicts/war? See: https://sdgs.un.org/goals/goal16. • Is sustainable economic growth possible? In what ways might the intervention of non-market entities such as the state, international organizations, and social forces, safeguard our planet? See: https://www.economicshelp.org/blog/145989/economics/environmental-impact-of-economic-growth/. • Might off-grid ecologies, such as natural and indigenous communities offer valuable insights, perspectives, and approaches to sustainable cities?

Appendix 1: Assessment Criteria for Written Report (Part 1 of the Group Project)

The written report is a team assessment, and accounts for 25% of your grade. Each team is to hand in a written report according to given instructions. The report should not be more than 2000 words with a 10% leeway. Sections not included in the word count: (i) cover page, abstract, content page, and references. In principle, all team members will receive the same mark. However, if the evaluation of team members' contribution to teamwork (see appendix 4) shows that the contribution of a team member is significantly different to that of the others, the individual grade may be adjusted to reflect that.

Written Report (25%)					
	Excellent	**Very Good**	**Average**	**Fair**	**Poor**
Practical Application and Feasibility (45%)	The project demonstrates a highly practical approach with clear, feasible applications. Solutions are well-supported with evidence and have significant potential for real-world implementation.	The project is practical with feasible applications, needing minor adjustments. Solutions are well-supported but might require some refinement.	The project shows some practicality and feasibility, but solutions need considerable improvement to be viable.	The project lacks practicality and feasibility, with solutions that are not well-supported or clearly defined.	The project is impractical and unfeasible, with no clear or supported solutions.
Creativity and Innovation (25%)	Extremely creative and innovative approach with clear, motivated objectives aligned with SDGs.	Fairly creative and innovative with well-defined objectives.	Some creativity with motivated objectives, but lacking clarity or coherence.	Struggles with creativity, weak motivation, and poor theme.	No creativity or motivation, undefined topic.

Integration of Three Systems' Perspectives (20%)	Critical discussion and natural integration of society, economy, and environment.	Meaningful consideration of all three systems.	Consideration of all systems to some degree but not interrelated.	Lacking coverage of one or two systems with no integration.	Lacking awareness of the systems' perspectives.
Clarity of Expression, Writing Style, Formatting, and Layout (10%)	Thorough knowledge, excellent and concise report, clear discussion, and critical insights.	Good knowledge, uniform layout and style, clear discussion, and minor errors.	Some knowledge, fair format, clear discussion, and some errors.	Poor knowledge, non-uniform layout, unclear discussion, and many errors.	Bad layout and clarity, poor content, and unclear figures.

*Interrelations: connections, contradictions, opposing or mutual interests.

Appendix 2: Assessment Criteria for Video Presentation (Part 2 of the Group Project)

The video presentation is a team assessment and accounts for 25% of your grades. For this assessment, you are to record a 12-15 min (maximum 15 mins) team presentation on video and submit it to your instructor through NTU Learn. Other students will watch your video presentation and give qualitative feedback through the Q&A session.

In principle, all team members will receive the same marks. However, if the evaluation of team members' contribution to teamwork (see appendix 3) shows that the contribution of a team member is significantly different to that of the others, the individual grade may be adjusted to reflect that.

Video Presentation on group project (25%)					
Criteria	**Excellent**	**Very Good**	**Average**	**Fair**	**Poor**

Practical Application and Feasibility (20%)	The project demonstrates a highly practical approach with clear, feasible applications and significant real-world potential.	Practical with feasible applications, needing minor adjustments.	Some practicality and feasibility, but solutions need considerable improvement.	Lacks practicality and feasibility, with unclear or unsupported solutions.	Impractical and unfeasible, with no clear or supported solutions.
Creativity and Innovation (20%)	Extremely creative with clear, motivated objectives.	Fairly creative with well-defined objectives.	Some creativity with motivated objectives, lacking clarity or coherence.	Struggles with creativity, weak motivation, and poor theme.	No creativity or motivation, undefined topic.
Integration of Three Systems' Perspectives (10%)	Critical discussion and natural integration of society, economy, and environment.	Meaningful consideration of all three systems.	Consideration of all systems to some degree but not interrelated.	Lacking coverage or integration.	Lacking awareness of the systems' perspectives.
Clarity of Expression, Organization, and Presenting Style (10%)	Clear, concise, well-structured, and engaging presentation with smooth delivery.	Mostly clear and well-structured with smooth delivery.	Partly clear presentation with occasional engagement.	Confusing and unattractive with expressionless delivery.	Failure to deliver effectively.
Answering questions (20%)	All members could answer questions with excellent and knowledgeable answers.	Few members show the ability to answer questions with good answers.	Only a few members attempt to answer questions.	Could not answer most of the questions.	Not able to answer any of the questions.

Raising questions **(20%)**	Demonstrates excellent preparation: questions display an excellent understanding of the topic and underlying concepts of sustainability. Questions also actively stimulate and sustain further discussion by building on peers' responses including building a focused argument around a specific issue, asking a newly related question, and making an oppositional statement supported by related research.	Demonstrates good preparation: questions display a good understanding of the topics and contribute to the ongoing conversations by asking a related question or making an oppositional statement supported by related research	Demonstrates adequate preparation: questions show adequate understanding of the topic and only repeat or summarize basic information without considering alternative perspectives or connections between ideas based on relevant research.	Demonstrates little or no evidence of preparation: questions are largely personal opinions or feelings without supporting statements; showing little or no evidence from watching other group's video presentation or relevant research.	Not able to raise any questions

Appendix 3: Peer review (Teamwork Evaluation)

1. All team members are expected to complete the teamwork evaluation for all other members in the team (i.e., self-assessment is not required); see evaluation criteria on the course outline. *Should a student fail to complete the teamwork evaluation of all other*

members in his/her group, <u>5% of the total grade will be deducted</u> from the student's final team assignment mark.

2. All assessments and qualitative comments are *confidential*. Notably, t<u>eam members will be able to see the compiled **quantitative and qualitative comments** given by their team members</u>.
   Note: *Students should avoid identifying themselves in any manner when providing qualitative comments to their team members*.

3. Each member's average rating given by his/her team members will be used to determine the final team assignment marks awarded to each member as follows:

Average Rating Range (min: 1; max: 9)	**Marks Deduction (out of 100%)**	**Final Adjusted Marks (out of 100%)**
≥ 6.5	0%	Original team assignment mark (**M%**)
≥ 5.5 to < 6.5	-5%	**M%** – 5%
≥ 4.0 to < 5.5	-10%	**M%** – 10%
≥ 3.0 to < 4.0	-15%	**M%** – 15%
> 1.0 to < 3.0	-20%	**M%** – 20%
= 1.0	**-M%**	0%

4. Please complete the teamwork evaluation through Peerceptiv for all your team members for the team assignment at the specified course dates.

5. If you have any concern with any of the quantitative ratings and qualitative comments that you receive from other members in your team, ***please consult your course instructor <u>within one week</u>*** upon receiving the ratings / qualitative comments.

6. Please note that the teaching team reserves the right to adjust students' final team assignment marks based on additional considerations including gathered information, certified special education needs (SEN), and medical diagnosis.

NOTES:

Please note that the teaching team reserves the right to adjust students' final team assignment marks based on additional considerations including gathered information, certified special education needs (SEN), and medical diagnosis.